Data Analysis Plan

Introduction: The Framingham Heart Study is a long-term prospective cohort initiated in 1948 to investigate cardiovascular disease risk in a community-based population. This analysis uses a subset of 4,434 participants with baseline measurements of cardiovascular risk factors and stroke outcomes. Although participants were followed for up to 24 years, this study focuses on the first 10 years of follow-up to ensure that baseline risk factors remain relevant to the outcome window. The primary objective is to identify baseline risk factors associated with time to first stroke within 10 years, analyzed separately for males and females. A secondary objective is to estimate 10-year stroke probabilities for clinically meaningful risk profiles. Additionally, systolic blood pressure and diabetes will be summarized across examination periods to assess temporal changes and motivate future analyses incorporating time-varying covariates.

Methods

Data Cleaning: The analytic cohort was restricted to Period 1 (baseline) observations among participants free of prior stroke (n = 4,434). Covariates include age, systolic blood pressure, BMI, total cholesterol, diabetes status, smoking status, antihypertensive medication use, and coronary heart disease. A complete case analysis was performed (n = 4,275 for the analytic sample). Follow-up time was measured from baseline and capped at $365.25 \times 10 = 3{,}652.5$ days, using 365.25 days per year to account for leap years, to restrict the analysis to 10 years. Participants experiencing non-stroke events or remaining event-free beyond 10 years were censored. Biologically implausible values will be reviewed and excluded.

Data Analysis: Cox proportional hazards models will be used to assess associations be-

tween baseline covariates and time to incident stroke. Kaplan–Meier curves and log-rank tests will be used for preliminary comparisons of stroke-free survival. Variable selection will be performed using backward selection or LASSO, while retaining age, systolic blood pressure, and diabetes in all models. To facilitate interpretation, the same set of covariates will be used for male and female models, with selection informed by results from either group. Variable selection will precede handling of missing data, and final models will be fit using complete-case analysis restricted to selected covariates to maximize data use. Non-stroke events will be treated as censored, and competing risks will not be explicitly modeled. The proportional hazards assumption will be assessed using Schoenfeld residuals and graphical diagnostics. Based on the final models, 10-year survival probabilities will be estimated for individuals aged 40, 50, and 60 under five scenarios: (1) baseline risk profile, (2) high systolic blood pressure (160 mmHg), (3) diabetes, (4) both high blood pressure and diabetes, and (5) an additional profile selected based on model results. Results will be presented separately for males and females, with emphasis on within-sex interpretation. To inform future analyses, systolic blood pressure and diabetes prevalence will be summarized descriptively across examination periods by sex to assess potential time-varying effects.

Preliminary Descriptive Analysis: Baseline characteristics were summarized in Table 1 stratified by sex. with continuous variables reported as mean (SD) and median [min, max] and categorical variables as count (%). Exploratory figures include a boxplot of systolic blood pressure by antihypertensive medication use (Figure 1), histograms of systolic blood pressure by sex (Figure 2), and a scatterplot of age versus systolic blood pressure with a loess smoother (Figure 3). All analyses were conducted using R.

Table 1: Baseline Characteristics by Sex

	Male	Female	Overall
	(N=1897)	(N=2378)	(N=4275)
Age (years)			
Mean (SD)	49.7 (8.71)	49.9 (8.59)	49.8 (8.64)
Median [Min, Max]	49.0 [33.0, 69.0]	49.0 [32.0, 70.0]	49.0 [32.0, 70.0]
Systolic BP (mmHg)			
Mean (SD)	132 (19.3)	134 (24.3)	133 (22.2)
Median [Min, Max]	129 [83.5, 232]	129 [83.5, 295]	129 [83.5, 295]
BMI (kg/m²)			
Mean (SD)	26.2 (3.40)	25.6 (4.52)	25.8 (4.07)
Median [Min, Max]	26.1 [15.5, 40.4]	24.8 [16.0, 56.8]	25.5 [15.5, 56.8]
Total Cholesterol (mg/dL)			
Mean (SD)	234 (42.5)	240 (46.1)	237 (44.6)
Median [Min, Max]	231 [113, 696]	237 [129, 600]	234 [113, 696]
Coronary Heart Disease			
No	1784 (94.0%)	2314 (97.3%)	4098 (95.9%)
Yes	113 (6.0%)	64 (2.7%)	177 (4.1%)
Diabetes			
No	1841 (97.0%)	2321 (97.6%)	4162 (97.4%)
Yes	56 (3.0%)	57 (2.4%)	113 (2.6%)
Current Smoker			
No	750 (39.5%)	1420 (59.7%)	2170 (50.8%)
Yes	1147 (60.5%)	958 (40.3%)	2105 (49.2%)
Antihypertensive Medication			
No	1857 (97.9%)	2286 (96.1%)	4143 (96.9%)
Yes	40 (2.1%)	92 (3.9%)	132 (3.1%)
Incident Stroke within 10 Years			
No	1849 (97.5%)	2321 (97.6%)	4170 (97.5%)
Yes	48 (2.5%)	57 (2.4%)	105 (2.5%)

Figure 1. Baseline Systolic Blood Pressure by Antihypertensive Medication

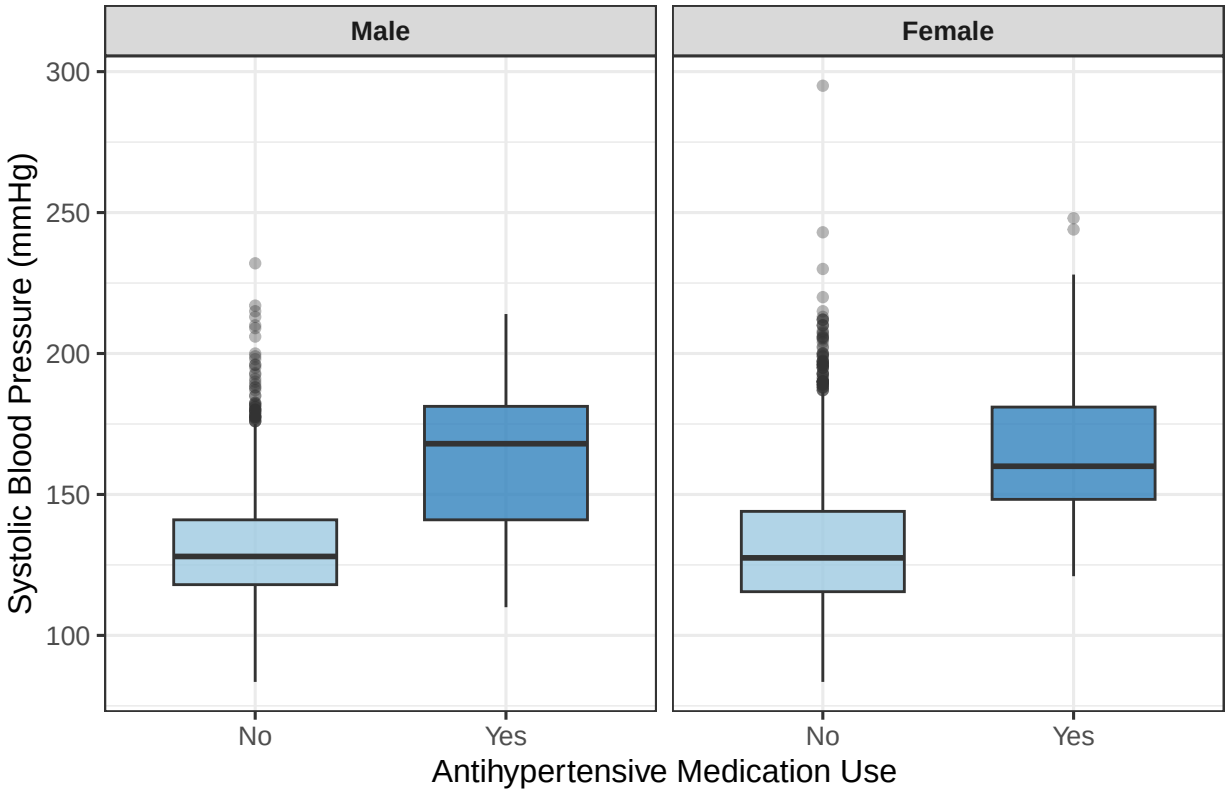

Figure 2. Distribution of Baseline Systolic Blood Pressure by Sex

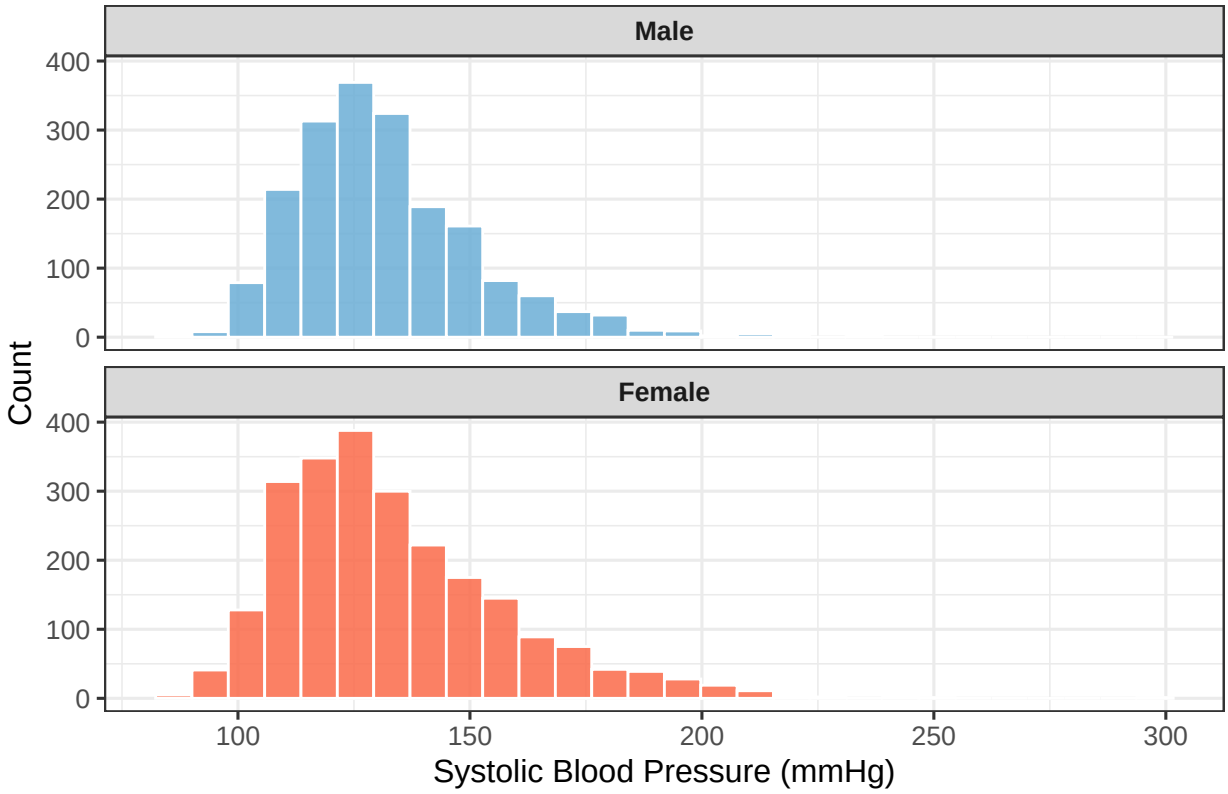
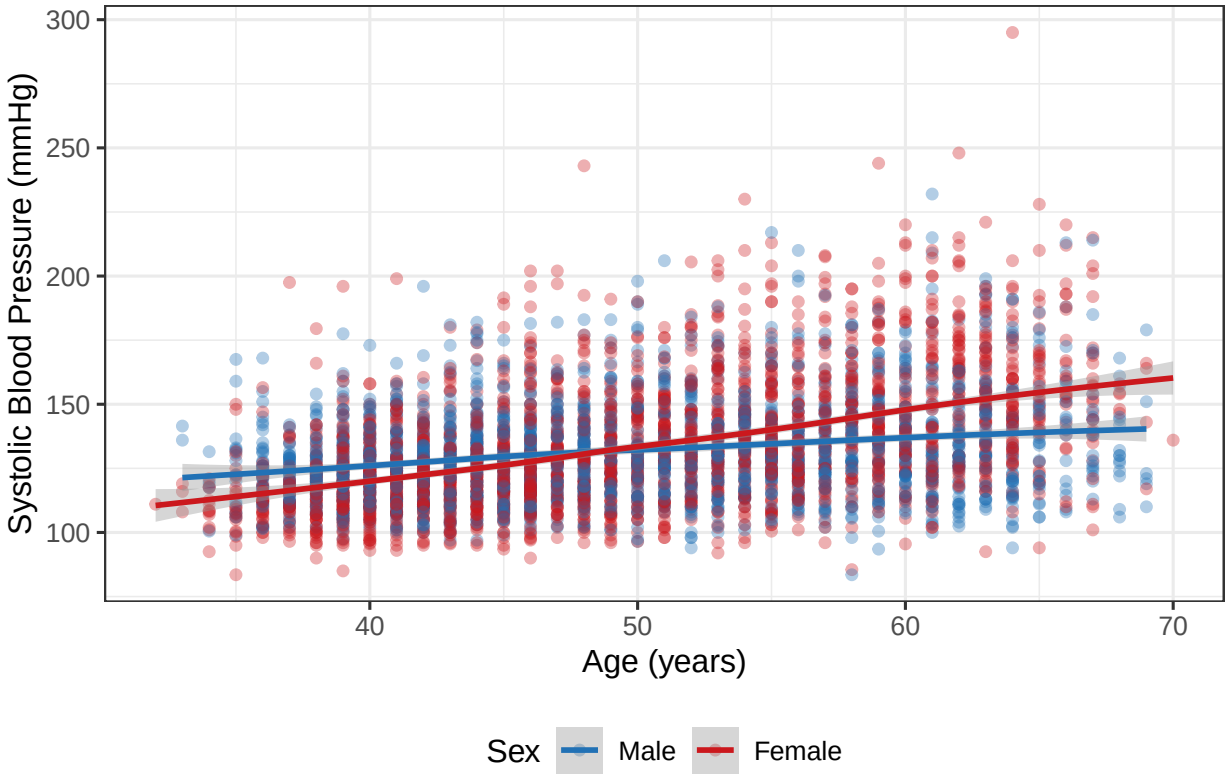
Figure 3. Relationship Between Age and Baseline Systolic Blood Pressu